

UNIT-II SERVER AND DESKTOP VIRTUALIZATION

2.Virtual Machines:

Virtual machines (VMs) are a fundamental technology in modern computing, offering a way to create isolated computing environments on a single physical machine.

Essentially, a VM acts as a virtual computer with its own dedicated resources, including:

- **Operating System:** A VM can run its own operating system (OS) independent of the host machine's OS. This allows users to run applications designed for different operating systems on a single machine.
- **Hardware Resources:** VMs are allocated a specific amount of processing power (CPU), memory (RAM), and storage from the physical machine's resources. This isolation ensures that applications running within a VM do not interfere with other VMs or the host machine itself.

Benefits of Virtualization:

- **Improved Resource Utilization:** VMs allow for efficient utilization of hardware resources by consolidating multiple workloads onto a single physical machine. This reduces hardware costs and energy consumption.
- **Increased Agility and Scalability:** VMs can be quickly provisioned, deployed, and migrated, enabling rapid scaling of computing resources to meet changing demands.
- **Enhanced Security and Isolation:** Each VM operates in a secure sandbox, preventing applications within a VM from impacting other VMs or the host system. This improves overall system stability and security.
- **Simplified Testing and Development:** VMs provide isolated environments for testing and developing software, allowing developers to experiment without risking the stability of the production environment.

Applications of Virtual Machines:

- **Server Consolidation:** Businesses can consolidate multiple physical servers onto fewer machines using VMs, reducing hardware costs and simplifying management.
- **Desktop Virtualization:** VMs can be used to deliver virtual desktops to users, offering a centralized and secure way to access applications and data from any device.

- **Software Development and Testing:** Developers can create isolated environments for testing and development to ensure software compatibility and functionality.
- **Cloud Computing:** VMs are the foundation of cloud computing, providing on-demand access to computing resources like servers, storage, and applications.

In conclusion, virtual machines offer a powerful and versatile tool for maximizing computing efficiency, enhancing security, and enabling flexible resource allocation. Their diverse applications make them a cornerstone of modern IT infrastructure, facilitating cloud computing, software development, and server consolidation.

Virtual machine defined

A VM is a virtualized instance of a computer that can perform almost all of the same functions as a computer, including running applications and operating systems. Virtual machines run on a physical machine and access computing resources from software called a hypervisor. The hypervisor abstracts the physical machine's resources into a pool that can be provisioned and distributed as needed, enabling multiple VMs to run on a single physical machine.

How multiple virtual machines work

Multiple VMs can be hosted on a single physical machine, often a server, and then managed using virtual machine software. This provides flexibility for compute resources (compute, storage, network) to be distributed among VMs as needed, increasing overall efficiency. This architecture provides the basic building blocks for the advanced virtualized resources we use today, including cloud computing.

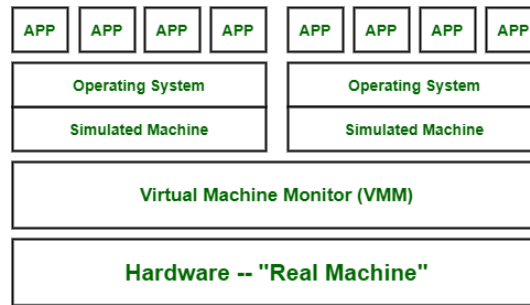
2.1 Types of Virtual Machines

1. System virtual machines

These kinds of VMs are completely virtualized to replace a real machine. The way they virtualize depends on a hypervisor such as VMware ESXi, which can operate on an operating system or bare hardware.

The hardware resources of the host can be shared and managed by more than one virtual machine. This makes it possible to create more than one environment on the host system. Even though these environments are on the same physical host, they are kept separate. This lets several single-tasking operating systems share resources concurrently.

System Virtual Machine



Different VMs on a single computer operating system can share memories by applying memory overcommitment systems. This way, users can share memory pages with identical content among multiple virtual machines on the same host, which is helpful, especially for read- only pages.

Advantages of system VMs are:

A System virtual machines have the capability, either via emulators or by using just-in- time compilation, of providing a simulated hardware environment. This environment is distinct from the instruction set architecture of the host (ISA).

The virtual machine software users choose comes packed with application provisioning that allows users to create packages, high availability, maintenance, and disaster recovery. This makes the tools for virtual machines more straightforward to use, making it possible for many operating systems to operate effectively on a single host.

The presence of a virtual partition allows for multiple OS environments to co-exist on the same primary drive. This partition allows for sharing files generated from the host or the guest operating environment. Other processes, such as software installations, wireless connections, and remote replications, such as printing, can be performed efficiently in the host's or guest's environment.

It allows developers to perform tasks without changing operating systems. All the generated data is stored on the host's hard drive.

Disadvantages of system virtual machines are:

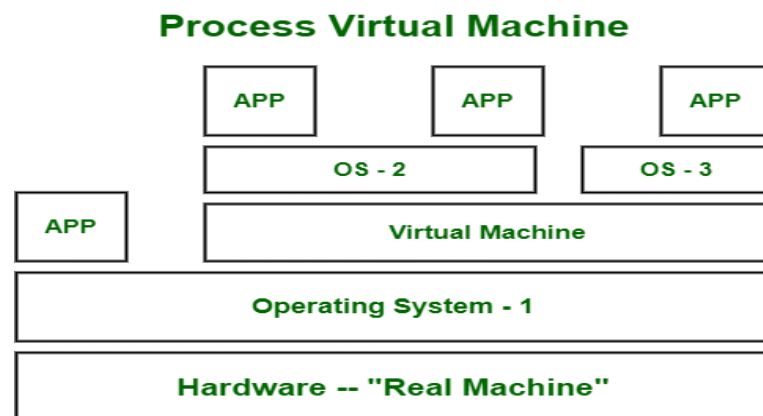
When virtual machines indirectly access the host's hard drive, they become less efficient than actual machines. Depending on the system, the performance of several virtual machines running on the same host can be different. The speed of execution and the protection against malware can also vary, leading to unstable behaviour. Users can mitigate this problem using virtual machine software that provides temporal isolation.

The guest operating system may not necessarily be compatible with the malware protections provided by the host resources. Therefore, it may require additional separate software leading to increased costs.

Process virtual machines

These virtual machines are sometimes called application virtual machines or Managed Runtime Environments (MREs). They run as standard applications inside the host's operating system, supporting a single process. It is triggered to launch when the process starts and destroyed when it exits. It offers a platform-independent programming environment to the process, allowing it to execute similarly on any platform.

The Process virtual machines are implemented using interpreters and they provide high- level abstractions. They are often used with Java programming language, which uses Java virtual machines to execute programs. There can be two more examples of process VMs i.e., The Parrot virtual machine and the .NET Framework that runs on the Common Language Runtime VM. Additionally, they operate as an abstraction layer for any computer language being used.



A process virtual machine may, under some circumstances, take on the role of an abstraction layer between its users and the underlying communication mechanisms of a computer cluster. In place of a single process, such a virtual machine (VM) for a process consists of one method for each real computer that is part of the cluster.

A Special case process VMs enable programmers to concentrate on the algorithm instead of the communication process provided by the virtual machine OS and the interconnect.

These VMs are based on an existing language, so they don't come with a specific programming language. Their systems provide bindings for several programming languages, such as Fortran and C. In contrast to other process VMs, they can enter all OS services and aren't limited by the system model. Therefore, it cannot be categorized strictly as virtual machines.

2.2 Server Virtualization

Server Virtualization is the process of dividing a physical server into several virtual servers, called virtual private servers. Each virtual private server can run independently. The concept of Server Virtualization widely used in the IT infrastructure to minimizes the costs by

increasing the utilization of existing resources.

Types of Server Virtualization

1. Hypervisor

In the Server Virtualization, Hypervisor plays an important role. It is a layer between the operating system (OS) and hardware. There are two types of hypervisors.

🚦 Type 1 hypervisor (Bare metal or native hypervisors) 🚦 Type 2 hypervisor (Hosted or Embedded hypervisors)

The hypervisor is mainly used to perform various tasks such as allocate physical hardware resources to several smaller independent virtual machines, called "guest" on the host machine.

2. Full Virtualization

Full Virtualization uses a hypervisor to directly communicate with the CPU and physical server. It provides the best isolation and security mechanism to the virtual machines. The biggest disadvantage of using hypervisor in full virtualization is that a hypervisor has its own processing needs, so it can slow down the application and server performance. VMWare ESX server is the best example of full virtualization.

3. Para Virtualization

Para Virtualization is quite similar to the Full Virtualization. The advantage of using this virtualization is that it is easier to use, Enhanced performance, and does not require emulation overhead. Xen primarily and UML use the Para Virtualization. The difference between full and para virtualization is that, In para virtualization hypervisor does not need too much processing power to manage the OS.

4. Operating System Virtualization

The Operating system virtualization is also called as system-level virtualization. It is a server virtualization technology that divides one operating system into multiple isolated user-spaces called virtual environments. The biggest advantage of using server visualization is that it reduces the use of physical space, so it will save money. Linux OS Virtualization and Windows OS Virtualization are the types of Operating System virtualization. FreeVPS, OpenVZ, and Linux Vserver are some examples of System-Level Virtualization.

5. Hardware Assisted Virtualization

Hardware Assisted Virtualization was presented by AMD and Intel. It is also known

as Hardware virtualization, AMD virtualization, and Intel virtualization. It is designed to increase the performance of the processor. The advantage of using Hardware Assisted Virtualization is that it requires less hypervisor overhead.

6. Kernel-Level Virtualization

Kernel-level virtualization is one of the most important types of server virtualization. It is an open-source virtualization which uses the Linux kernel as a hypervisor. The advantage of using kernel virtualization is that it does not require any special administrative software and has very less overhead. User Mode Linux (UML) and Kernel-based virtual machine are some examples of kernel virtualization.

Advantages of Server Virtualization

1. Independent Restart

In Server Virtualization, each server can be restart independently and does not affect the working of other virtual servers.

2. Low Cost

Server Virtualization can divide a single server into multiple virtual private servers, so it reduces the cost of hardware components.

3. Disaster Recovery

Disaster Recovery is one of the best advantages of Server Virtualization. In Server Virtualization, data can easily and quickly move from one server to another and these data can be stored and retrieved from anywhere.

4. Faster deployment of resources

Server virtualization allows us to deploy our resources in a simpler and faster way.

5. Security

It allows users to store their sensitive data inside the data centres.

Disadvantages of Server Virtualization

- ✚ The biggest disadvantage of server virtualization is that when the server goes offline, all the websites that are hosted by the server will also go down.
- ✚ There is no way to measure the performance of virtualized environments. ✚ It requires a huge amount of RAM consumption.
- ✚ It is difficult to set up and maintain.

Some core applications and databases are not supported virtualization.

- ✚ It requires extra hardware resources. Uses of Server Virtualization.

- ✚ Server Virtualization is used in the testing and development environment. ✚ It improves the availability of servers.

- ✚ It allows organizations to make efficient use of resources.

- ✚ It reduces redundancy without purchasing additional hardware components.

2.3 Business Cases of Server Virtualization

Business challenge: We have many stand-alone HMI workstations running different versions of the HMI client application and are looking for a simpler and more cost effective approach to managing our applications and client operating systems.

Business solution: Investment in virtualization technology and hardware that would provide the foundation for future operations for managing the application and client hardware infrastructure.

Business benefits: Allowed customer to introduce the use of zero clients running ACP thin manager to allow for device management running on a server based virtual infrastructure in which application deliver was also made simpler through the use of server-based deployment via Citrix/RDS.

Business Benefits of Server Virtualization

Virtualization is not just an IT trend. It is also not new, but it is new in many organizations, as companies of all sizes invest in virtualization technologies to reap it's many benefits: server and desktop provisioning, reduction in physical servers, increased uptime and availability, better disaster recovery, energy savings...and the list goes on.

Switching to virtualization means that the workloads happening on servers are not tied to a specific piece of physical hardware and that multiple virtual workloads can occur simultaneously on the same piece of machinery. The immediate benefits of virtualization include higher server utilization rates in the and lower costs, but there are more sophisticated advantages as well.

Server virtualization has proven itself to be a revolutionary technology solution for IT management, presenting capabilities that would never be possible within a physical infrastructure. At an economic stance, the benefits of server virtualization are focused on cost savings because it allows multiple applications to be installed on a single physical server. So how exactly can your company benefit from a virtual infrastructure? Let's take a look at the top five ways:

1. Reduced Hardware Costs

It is said that humans theoretically only use 10% of their brain command; most of the servers in a strictly physical environment are heavily under-utilized, using an estimated 5-15% of their capacity. When you implement a virtualized server / cloud computing approach, hardware utilization is increased because one physical server can now hold multiple virtual machines. Applications no longer need their own server because each virtual machine on the physical server now runs them. In 2011, IDC reported a 40% reduction in hardware and software costs for IT departments that adopted a server virtualization strategy.

2. Faster Server Provisioning and Deployment

Server virtualization enables system provisioning and deployment within minutes, allowing you to clone an existing virtual machine without the hours and costs normally spent procuring and installing a new physical server. Companies with virtual environments already look back and cringe at the grueling process of filling out a purchase order, waiting for the server to arrive and then waiting hours for the operating system and applications to finish installing.

Time and cost add up substantially, not to mention the growing number of racks and cables you would have to purchase to accommodate for the increasing number of physical servers. Virtualization is most certainly necessary for most businesses to keep up with the explosion of data resources needed to keep pace with competitors.

3. Greatly Improved Disaster Recovery

Perhaps the greatest benefit of server virtualization is the capability to move a virtual machine from one server to another quickly and safely. Backing up critical data is done quickly and effectively because your company can effortlessly create a replication site. Most enterprise virtualization platforms contain software that helps automate the failover during a disaster. The software also allows you to test a disaster recovery failover think of it as your data center's own fire escape plan. If a data center disaster occurs, your infrastructure is already set up to take appropriate measures for a swift and safe recovery. Try achieving that with arrays of physical servers now that's a real disaster.

4. Increased Productivity

Few physical servers' means there are less of them to maintain and manage. A benefit of applications that used to take days or weeks to provision are now done in minutes. This leaves your IT/OT staff more time to spend on more productive tasks such as driving new business initiatives, cutting expenses and raising revenue.

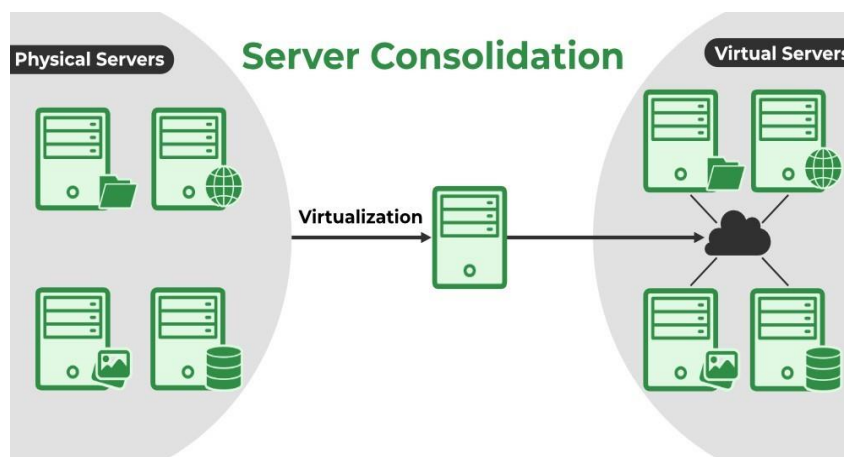
5. Energy Cost Savings

Among other server virtualization benefits, the migration of physical servers to virtual machines allows you to consolidate them onto fewer physical servers. The result? Cooling and power costs are significantly reduced, which means not only will you be “going green,” but you will also have more green to spend elsewhere. According to VMware, server consolidation reduces energy costs by up to 80%. Another major plus is the ability to power down servers without affecting applications or users.

Server virtualization brings positive transformations, such as reduced hardware costs, improved server provisioning and deployment, better disaster recovery solutions, efficient and economic use of energy, and increased staff productivity. Still, it may seem like a daunting task to move to a virtual infrastructure.

2.4 Uses of virtual server consolidation

A Server consolidation in cloud computing refers to the process of combining multiple servers into a single, more powerful server or cluster of servers. This can be done in order to improve the efficiency and cost-effectiveness of the cloud computing environment. Server consolidation is typically achieved through the use of virtualization technology, which allows multiple virtual servers to run on a single physical server. This allows for better utilization of resources, as well as improved scalability and flexibility. It also allows organizations to reduce the number of physical servers they need to maintain, which can lead to cost savings on hardware, power, and cooling.



Benefits of Server Consolidation

Cost savings: By consolidating servers, organizations can reduce the number of physical servers they need to maintain, which can lead to cost savings on hardware, power, and cooling.

Improved performance: Consolidating servers can also improve the performance of the cloud computing environment. By using virtualization technology, multiple virtual servers can run on a single physical server, which allows for better utilization of resources. This can lead to faster processing times and better overall performance.

Scalability and flexibility: Server consolidation can also improve the scalability and flexibility of the cloud environment. By using virtualization technology, organizations can easily add or remove virtual servers as needed, which allows them to more easily adjust to changing business needs.

Management simplicity: Managing multiple servers can be complex and time-consuming. Consolidating servers can help to reduce the complexity of managing multiple servers, by providing a single point of management. This can help organizations to reduce the effort and costs associated with managing multiple servers.

Better utilization of resources: By consolidating servers, organizations can improve the utilization of resources, which can lead to better performance and cost savings. Server consolidation in cloud computing is a process of combining multiple servers into a single, more powerful server or cluster of servers, in order to improve the efficiency and cost-effectiveness of the cloud computing environment.

Virtualization Platform

A big part of a virtualized system's success depends on implementation architecture choices. Here we'll take a look at how to choose a virtualization platform.

Hardware Environment

These are the servers, storage and networking components of the data center being virtualized. It is possible to reuse the hardware already in place. In fact, virtualization supports this by providing a consistent interface to the application to be deployed even if the hardware differs. The hardware environment chosen plays a crucial role in determining the software platform to be used.

Software Platform

The software layer abstracts the hardware environment to provide the hosted environments with an idealized environment. Distinct virtualization software has unique hardware requirements so when existing hardware is to be used, software choices are limited by compatibility with the hardware. Even when compatible hardware is used, the specifics influence performance. If exceptionally high performance is critical, hardware should be chosen very carefully.

2.5 Virtualization Platforms

VMware's vSphere is the celebrity in virtualization's climate-controlled data room, but it is not the only option. There are other platforms available for virtualization, including XenSource and Hyper-V, which are stable, enterprise-ready products backed by established players and that provide long-term support.

The platform to use should not be decided based on cost alone, as sub-optimal solutions will eventually necessitate additional expenditures, resulting in higher long-term costs. There is no best solution here as finding the right fit involves individual considerations. That said, there are a few pointers when it comes to choosing the best possible virtualization platform.

Features to Look for in a Virtualization Platform

Reduction of Capital Expenditures (CAPEX): Typically servers have low utilization levels averaging around 15 percent. Virtualization can increase throughput four-fold. This means that a company can use less hardware and reduce energy costs. Note that this must be weighed against the total cost of ownership of the virtualization platform.

System Consolidation: By running multiple applications and operating systems, an organization reduces the number of physical servers required.

Server Consolidation: This is an approach to the efficient usage of computer server resources in order to reduce the total number of servers or server locations that an organization requires.

Ease of System Management: Easy management is how rapidly new services such as platform as service (PaaS), infrastructure as a service (IaaS) and software as a service (SaaS) can be deployed. It also refers to agility and speed in deploying new application stacks.

Platform Maturity: Virtualization is an expensive long-term investment which will be wasted if vendors are switched. As such, new entrants to the market may not be a good option for mission-critical data centers.

Hardware Compatibility: Virtualizing platforms have issues with hardware, including outright incompatibility and sub-optimal performance. Hosted hypervisors tend to have the greatest support for a range of hardware types, while bare-bones virtualizers are typically choosy about supported hardware. If you are planning to repurpose existing hardware, you should check if your platform of choice supports that hardware.

Total Cost of Ownership: Total cost of ownership is not just the price of purchasing a virtualization platform. Additional expenses such as guest OS licenses, training and support plus hardware should be factored in. One cost component that is often hidden is the purchase of advanced management tools. As a rule, these are not included in the initial purchase price. For instance, if you go for VMware you are likely to need vCenter Server as the installation scales up while System Center Virtual Machine Manager is the comparable offering from Microsoft to go with its Hyper-V. It might also be necessary to invest in third-party tools for auditing, capacity planning or reporting.

Technical Differences: When virtualizing, it is also important to consider technical concerns early on in the process in order to fully realize the benefits mentioned above without compromising efficiency. There are two main approaches to virtualizing:

Type 1 Hypervisor: Bare-metal virtualization, a hypervisor is the first layer of software on a clean x86 system. This provides more efficiency, performance, stability and greater scalability than hosted architectures. vSphere is a good example of this approach.

Type 2 Hypervisor: Hosted virtualization, virtualization is implemented on top of a standard operating system and supports the widest range of hardware. Microsoft's Hyper-V is the best example of this approach.

The top virtualization platforms are more or less evenly matched as far as features are concerned. However, features are not the only consideration when making hypervisor choice. Some of the other important factors are hardware compatibility and cost of ownership. System support, stability and scalability deserve a long hard look as well. The key is to assess all of a platform's features and costs against your company's needs and budget. The key isn't just to virtualize. For true success, companies must find the right fit

2.6 What is desktop virtualization

Desktop virtualization includes the replacement of traditional physical desktop computing environments with virtual computing environments. It helps create and store multiple user desktop environments on a single host, residing in the cloud or a data center.

Desktop virtualization is the one-stop solution to many problems that organizations face today, including regulatory compliance, security, cost control, business continuity and manageability.

Types of desktop virtualization

1. Virtual Desktop Infrastructure

A virtual desktop interface (VDI) uses host-based virtual machines (VMs) to run the operating system. It delivers non-persistent and persistent virtual desktops to all connected devices. With a non-persistent virtual desktop, employees can access a virtual desktop from a shared pool, whereas in a persistent virtual desktop, each user gets a unique desktop image that can be customized with data and applications. VDI gives each user their virtual machine and supports only one user per operating system.

2. Remote Desktop Services

Remote desktop services (RDS) or remote desktop session host (RDSH) are beneficial where only limited applications require virtualization. They allow users to remotely access Windows applications and desktops using the Microsoft Windows Server operating system. RDS is a more cost-effective solution, since one Windows server can support multiple users.

3. Desktop-as-a-Service (DaaS)

Desktop-as-a-service (DaaS) is a flexible desktop virtualization solution that uses

cloud- based virtual machines backed by a third-party provider. Using DaaS, organizations can outsource desktop virtualization solutions that help a user to access computer applications and desktops from any endpoint platform or device.

To understand which type of solution suits your business, you should:

Identify the costs associated with setting up the infrastructure and deployment of virtual desktops
Determine whether you have the required resources and expertise to adopt these solutions
Determine the infrastructure control capabilities of the virtualization providers
Determine the level of elasticity and agility you want in your desktop virtualization solution

Why do you need desktop virtualization for your business.

Beyond saving money and time, desktop virtualization offers various other benefits for organizations. These include:

Better security and control: Virtual desktops store data in a secure environment and allow central management of confidential information to prevent data leaks. Desktop virtualization solutions restrict the users from saving or copying data to any source other than its servers, making it hard to get crucial company information out. Reliable virtualization solution providers offer multiple layers of cloud safeguards such as the highest quality encryption, switches, routers and constant monitoring to eliminate threats and protect users' data.

Ease of maintenance: Unlike traditional computers, virtual desktops are far easier to maintain. All end-users don't need to update or download the necessary programs individually since these are centrally managed by the IT department.

IT admin can also easily keep track of the software assets through a virtual desktop. Once a user logs off from the virtual desktop it can reset, and any customizations or software programs downloaded on the desktop can be easily removed. It also helps prevent system slowdown caused by customizations and software downloads.

Remote work: Since virtual desktops are connected to a central server, provision for new desktops can be made in minutes so they are instantly available to new users. Instead of manually setting up a new desktop for new employees, IT admins can deploy a ready-to-go virtual desktop to the new user's device using desktop virtualization. The users can access and interact with the operating systems and applications from virtually anywhere with an internet connection.

Resource management: Resources for desktop virtualization are located in a data center, which allows the pooling of resources for better efficiency. With desktop virtualization, IT admins can maximize their hardware investment returns by consolidating the majority of their computing in a data center. This helps eliminate the need to push application and operating system updates to the end-user machines.

Organizations can deploy less expensive and less powerful devices to end-users since

they are only used for input and output. IT departments can save money and resources that would otherwise be used to deploy more expensive and powerful machines.

Reduced costs: Desktop virtualization solutions help shift the IT budget from capital expenses to operating expenses. Organizations can prolong the shelf life of their traditional computers and other less powerful machines by delivering compute-intensive applications via VMs hosted on a data center.

IT departments can also significantly save costs on software licensing as you only need to install and update the software on a single, central server instead of multiple end-user workstations. Savings on energy bills, capital costs, licensing costs, IT support costs and upfront purchasing costs can reduce the overall IT operating costs by almost 70%.

Increased employee productivity: Employees' productivity may increase when they can easily access the organization's computing resources from any supported device, anywhere and anytime. Employees can work in a comfortable environment and still be able to access all applications and software programs that would otherwise only be available on their office

desktops. Desktop virtualization allows for a seamless and faster employee onboarding process, productivity and provisioning for remote workers.

Improved flexibility: Desktop virtualization reduces the need to configure desktops for each end-user. These solutions help organizations manage and customize desktops through a single interface and eliminate the need to personalize each desktop. Desktop virtualization lets the administrator set permissions for access to programs and files already stored on the central server with just a few clicks. Employees can access the required programs from anywhere, offering them better work flexibility.

